Sparse Poisson GLM with Off-Diagonal Memory Kernel: Model, Inference, and Training

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Abstract

Two-block inference scheme for a Poisson GLM with a lagged off-diagonal memory kernel on a non-stationary neuronal population. The model uses a diagonal lag-1 drive plus a shared history kernel on off-diagonal inputs. Alternating optimization yields tractable convex subproblems for the kernel (λ_2 smoothness) and network parameters (λ_3 off-diagonal sparsity).

Contents

1 Generative Model

Population of N neurons, T time bins of width Δt. Spike counts $Y_t \in \mathbb{N}_0^N$, $t = 0, \dots, T-1$.

Parameters. Diagonal: $A^{\text{diag}} \in \mathbb{R}^N$ with $(A^{\text{diag}})_i = (A_{\text{true}})_{ii}$. Off-diagonal: $A^{\text{off}} \in \mathbb{R}^{N\times N}$, zeros on diagonal. Kernel: $\kappa \in \mathbb{R}^M$, 0-based indexing, anchor $\kappa_0 = 1$, learnable entries $\kappa_1, \dots, \kappa_{M-1}$. Bias: $B \in \mathbb{R}^N$. Clipping threshold: $\eta_{\text{clip}} = 5$.

Off-diagonal history.

$$S_t \;=\; \sum_{\ell=0}^{M-1} \kappa_\ell \, Y_{t-1-\ell} \;\in\; \mathbb{R}^N, \qquad t-1-\ell \geq 0.$$

State equation.

$$\eta_{t,i} \;=\; (A^{\text{diag}})_i \, Y_{t-1,i} \;+\; \big(A^{\text{off}} S_t\big)_i \;+\; B_i,$$
$$\mu_{t,i} \;=\; \exp\!\big(\tilde{\eta}_{t,i}\big)\,\Delta t, \quad \tilde{\eta}_{t,i} \;=\; \min\big(\eta_{t,i},\, \eta_{\text{clip}}\big).$$

Observation model.

$$Y_{t,i} \sim \text{Poisson}(\mu_{t,i}), \quad i = 1, \dots, N,\; t = 0, \dots, T-1.$$

2 Objective and Identifiability

Log-likelihood. Dropping the constant $\log(Y_{t,i}!)$ terms:

$$\mathcal{L} = \sum_{i=1}^{N} \sum_{t=0}^{T-1} \left[Y_{t,i}\, \tilde{\eta}_{t,i} - \exp(\tilde{\eta}_{t,i})\,\Delta t \right].$$

Objective function.

$$\mathcal{J} \;=\; \underbrace{\sum_{t=2}^{T} \sum_{i=1}^{N} \left[\mu_{t,i} - Y_{t,i}\,(\tilde{\eta}_{t,i} + \log \Delta t)\right]}_{\text{Poisson NLL}} \;+\; \underbrace{\lambda_2 \sum_{\ell=2}^{M-1} \big(\kappa_\ell - \kappa_{\ell-1}\big)^2}_{\text{kernel smoothness (Block 1)}} \;+\; \underbrace{\lambda_3\, \overline{|A_{\text{off}}|}}_{\text{off-diagonal sparsity (Block 2)}}, \tag{1}$$

where

$$\overline{|A_{\text{off}}|} \;=\; \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i\neq j}}^{N} |A^{\text{off}}_{ij}|. \tag{2}$$

Identifiability. The product $A^{\text{off}}_{ij}\,\kappa_\ell$ has a scaling degeneracy, removed by the anchor $\kappa_0 = 1$. If clipping activates frequently, a censored Poisson likelihood should replace the standard one.

3 Inference: Two-Block Alternating Scheme

The bilinear structure $A^{\text{off}}_{ij}\,\kappa_\ell$ precludes joint convexity. We use block coordinate descent: each block is convex given the other fixed, guaranteeing a non-increasing objective sequence converging to a stationary point.

3.1 Initialization

Initialize A^{diag} and B from mean firing rates and lag-1 covariances. Initialize κ with a damped-oscillator profile or simply $\kappa_0 = 1$ with small subsequent entries.

3.2 Block 1 (kernel update)

Fix $(A^{\mathrm{diag}}, A^{\mathrm{off}}, B)$. Update $\kappa_1, \ldots, \kappa_{M-1}$ by minimizing $\mathcal{J}$ with the λ_2 smoothness penalty. The problem is convex in κ since S_t is linear in κ.

3.3 Block 2 (network update)

Fix κ. Update $(A^{\mathrm{diag}}, A^{\mathrm{off}}, B)$ by minimizing the Poisson NLL plus the λ_3 L1 penalty (2). Dale's law sign constraints on columns of A^{off} can be added as linear inequalities.

4 Training Samples

Input: spike counts $Y_t \in \mathbb{N}_0^N$, $t = 0, \ldots, T-1$. Let $M_{\mathrm{cut}} \geq 2$ be the history cutoff.

Sample construction. Each sample at output time $t_{\mathrm{out}} \in [M_{\mathrm{cut}},\, T-1]$:

$$X^{(t_{\mathrm{out}})} = \left\{ Y_{t_{\mathrm{out}}-1},\, Y_{t_{\mathrm{out}}-2},\, \ldots,\, Y_{t_{\mathrm{out}}-M_{\mathrm{cut}}} \right\} \in \mathbb{N}_0^{N \times M_{\mathrm{cut}}}, \qquad y^{(t_{\mathrm{out}})} = Y_{t_{\mathrm{out}}} \in \mathbb{N}_0^N.$$

Sampling strategy. Draw t_{out} densely from $[M_{\mathrm{cut}}, T-1]$ with stride s; optionally enforce a minimum separation to reduce autocorrelation. Total samples:

$$N_{\mathrm{samples}} \approx \left\lfloor \frac{T - M_{\mathrm{cut}}}{s} \right\rfloor.$$

For validation, sample windows within random non-overlapping blocks of length $L \gg M_{\mathrm{cut}}$ to obtain a temporally separated hold-out set.

5 Kernel Normalization

We use 0-based lag indexing $\ell \in \{0, \ldots, M-1\}$ with anchor $\kappa_0 = 1$. An alternative $\|\kappa\|_1 = 1$ constraint was tested but significantly reduced non-stationarity in simulations, so the ground truth κ_{true} does not obey it.

The λ_2 smoothness penalty imposes no parametric form on κ. If prior knowledge constrains the kernel shape, one can reparametrize κ with fewer parameters while keeping $\kappa_0 = 1$.

6 Implementation Notes

Sparsity and sign constraints. The λ_3 penalty (2) promotes sparsity; normalization by $N(N-1)$ makes the penalty strength network-size invariant. Column sign constraints (Dale's law) can be imposed if neuron types are known.

Regularization tuning. Start λ_2 small; increase if the kernel estimate oscillates. Tune λ_3 to match expected connectivity density (e.g., 5–20%).

Validation. Simulate spike trains from inferred $(A^{\text{diag}}, A^{\text{off}}, B, \kappa)$ and compare summary statistics (firing rates, ISI distributions, pairwise correlations, burst statistics) against data. Inspect residuals to diagnose model misspecification.

7 Conclusion

The formulation provides identifiable kernel learning via $\kappa_0 = 1$ anchoring and complementary regularization: λ_2 controls the temporal kernel shape and λ_3 enforces sparse connectivity. The two-block scheme keeps each subproblem convex, and the dense sampling strategy maximizes use of long spike-train recordings.